KEY RATE DURATION AS A MEASURE OF YIELD CURVE RISK

Learning Outcome

- define key rate duration and describe its use to measure price sensitivity of fixed-income instruments to benchmark yield curve changes

As we have seen, effective duration measures a bond's sensitivity to changes in the benchmark yield curve if all yields change by the same amount. Key rate duration provides further insight into a bond's sensitivity to *non-parallel* benchmark yield curve changes. **Key rate duration** (or partial duration) is a measure of a bond's sensitivity to a change in the benchmark yield at a specific maturity. Such a measure is important to isolate the price responses of bonds to changes in the rates of key maturities on the benchmark yield curve.

Key rate durations define a security's price sensitivity over a set of maturities along the yield curve, with the sum of key rate durations being equal to the effective duration, as shown in Equations 4 and 5:

$$KeyRateDur_k = -\frac{1}{PV} \times \frac{\Delta PV}{\Delta r_k}; \quad (4)$$

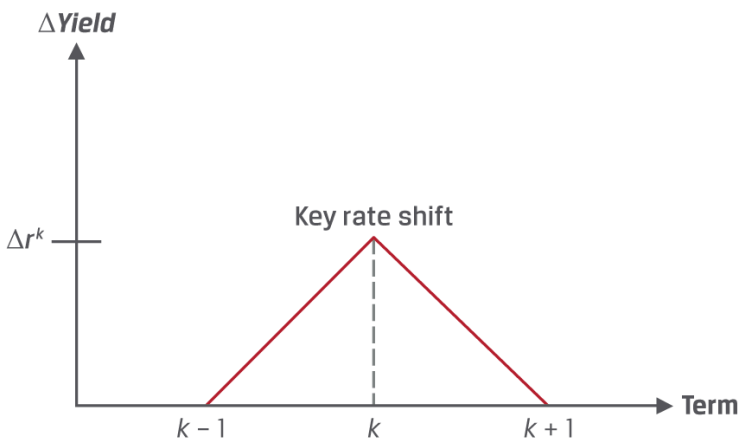
$$\sum_{k=1}^{n} KeyRateDur_k = EffDur, \quad (5)$$

where r_k represents the kth key rate. In contrast to effective duration, key rate durations help identify "shaping risk" for a bond—that is, a bond's sensitivity to changes in the shape of the benchmark yield curve (e.g., the yield curve becoming steeper or flatter or twisting).

The procedure for the calculation of key rate durations is similar to that used in the calculation of effective duration, but instead of shifting the entire benchmark yield curve, only key points on the curve are shifted, one at a time, as shown in Exhibit 3. The kth key rate, which could be a 0.5-, 2-, 5-, 10-, 20-, or 30-year rate, is shifted up (as shown in Exhibit 3) and down by 1 bp, and then new bond prices are generated (PV_+ and PV_-) and the key rate duration at that specific maturity is calculated using Equation 4. Thus, the effective duration for each maturity point shift is calculated in isolation.

Exhibit 3:

Key Rate Shift



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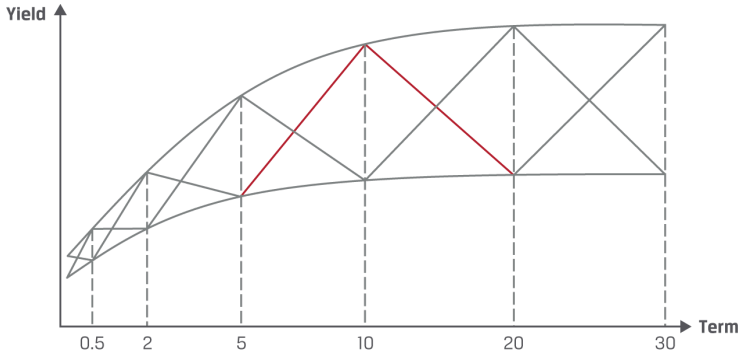
maturity.

Exhibit 4:

Key Rate Shifts across the Benchmark Yield Curve



As for practical uses of key rate duration, the analyst may want to know how the price of a callable bond is expected to change if short-term benchmark rates—for example, the rate on the two-year Treasury note, with modified duration of 1.99—rise by 25 bps but longer-maturity benchmark rates remain unchanged. Given that the yield curve is upward sloping, this scenario represents a flattening of the curve.

Equation 6 rearranges terms from Equation 4 to solve for $\Delta PV/PV$ (or $\%\Delta PV^{Full}$):

$$\frac{\Delta PV}{PV} = -KeyRateDur_k \times \Delta r_k. \qquad 6$$

Using Equation 6, the expected estimated price change would be -0.4975%:

$\Delta PV/PV = -1.99 \times 0.0025$.

Now, consider the simple portfolio of government bonds shown in Exhibit 5.

Exhibit 5:

Simple Government Bond Portfolio

Tenor (Years)	Coupon	Annualized Yield (%)	Price	Position Size	Modified Duration	Key Rate Duration
2	0.00	0.50	99.006	$99,006,219	1.990	0.711
5	0.00	1.25	93.960	$93,959,580	4.938	1.675
10	0.00	1.75	84.010	$84,009,625	9.828	2.981
				$276,975,424	5.368	5.368

Assume the portfolio is weighted by the prices of the respective 2-, 5-, and 10-year bonds for a total portfolio value of $277 million, or $1 million × (99.01 + 93.96 + 84.01). The portfolio's modified duration is calculated as

5.367 = [1.990 × (99.006/277.0)] + [4.938 × (98.3/277.0)] + [9.828 × (84.010/277.0)].

Alternatively, we could calculate each key rate duration by maturity. For example, the two-year key rate duration ($KeyRateDur_2$) is

0.711 = 1.990 × (99.006/277.0).

Note that the three key rate duration values sum to the portfolio duration value of 5.345.

If we assume this portfolio represents an "index," then a manager attempting to outperform the index could vary (or tilt) the key rate durations to capitalize on her view of expected

EXAMPLE 3

Continuing from Example 2, the portfolio manager asks you to determine if the bond portfolio may need rebalancing to improve overall return, given the tenor of the portfolio's bonds, key rate duration information, and forecasted benchmark government par curve shifts at the various maturities. All four bonds have the same benchmark government par curve. Current information for the bonds are as follows:

Bond	Tenor	Position Size	Key Rate Duration
A	1 year	$200,565,245	0.645
B	5 years	$201,042,132	1.483
C	10 years	$202,673,298	2.158
D	20 years	$202,588,801	2.982

Information on the forecasted benchmark government par curve changes at the various maturities are as follows:

Maturity	1 year	5 years	10 years	20 years
Expected change	–23 bps	–25 bps	–18 bps	–10 bps

If we take the product of the key rate duration and curve shift at a certain maturity, we can forecast the change in the price of the bond. For example, for the one-year maturity, this would be –0.645 × –0.0023 = 0.148%. The calculation for all the bonds is shown below:

Maturities	1 year (Bond A)	5 years (Bond B)	10 years (Bond C)	20 years (Bond D)
% change in bond price	0.148%	0.371%	0.388%	0.298%

Therefore, since the forecasted price change in the bonds favor Bonds B and C, it would be advisable to increase the weightings for these bonds, taking more from the Bond A position than the Bond D position, since Bond A is forecasted to give the least gain of all the bonds.

Question Set

Questions 1–3 relate to the following information.

The analysis you performed previously on the consideration of the impact of the benchmark government par curve increasing by 200 bps on two bonds has now been expanded for a more complicated scenario. You are asked to consider the scenario of a non-parallel shift in the benchmark government par curve, where a steepening of the curve is occurring at longer maturities. Below are the details on non-parallel shifts to the benchmark government par curve:

Maturity	1 year	5 years	10 years	20 years	30 years
Expected change	+100 bps	+150 bps	+200 bps	+250 bps	+300 bps

Relevant details on the two bonds are as follows:

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Bond B	10 years	3.953

1. Based on the information above, identify which bond will have a greater percentage change in its price due to the benchmark par curve shift.

Hide Solution

Solution

Bond B.

For Bond A, $\Delta PV/PV = -2.702 \times 0.015 = -4.05\%$.

For Bond B, $\Delta PV/PV = -3.953 \times 0.020 = -7.91\%$.

2. If the values in the table above on the non-parallel yield curve shifts were forecasts and your portfolio contains equal positions in the two bonds, which of the following would be the best course of action?

A. Sell Bond A.
B. Sell Bond B.
C. Sell both Bond A and Bond B.

Hide Solution

Solution

C is correct. Sell both Bonds A and B, since both are expected to see price declines.

3. Explain why key rate durations might be useful for a portfolio manager even though key rate duration values of the bonds in a portfolio sum to the portfolio duration.

Hide Solution

Solution

Knowing the portfolio duration and overall movement of the benchmark yield curve can provide a quick estimate of gains or losses; however, by using key rate durations, a portfolio manager can over- or underweight specific tenors to maximize risk-adjusted return.

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HA In exhibit 5 why did they use different values than that found in the table?

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PJ Is the Exhibit 5 just wrong

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